

# Prompting Techniques in Large Language Models

## 1 Introduction

Prompting is basically the way we communicate with a Large Language Model (LLM) to get the kind of response we want. Different prompting techniques can be used depending on whether we want to give examples, provide specific instructions, make the model take a particular role, or encourage it to solve a problem step by step.

## 2 Zero-Shot Prompting

In zero-shot prompting, we give the AI a task without providing any examples. The model has to understand the instruction and answer using the knowledge it already has.

### Prompt

Classify the following sentence as positive or negative:  
“The movie was amazing.”

### Output

Positive

The model was able to perform the task without being shown any example beforehand.

## 3 One-Shot Prompting

In one-shot prompting, we give the model one example before asking it to perform the actual task. The example helps the model understand what kind of output we expect.

## Prompt

Happy  $\rightarrow$  Positive  
Sad  $\rightarrow$  Negative  
Excited  $\rightarrow$  ?

## Output

Positive

The model uses the single example to understand the relationship between the input and output.

## 4 Few-Shot Prompting

In few-shot prompting, we provide multiple examples before asking the model to perform the actual task. This gives the model more information about the pattern we want.

## Prompt

Happy  $\rightarrow$  Positive  
Angry  $\rightarrow$  Negative  
Excited  $\rightarrow$  Positive  
Bored  $\rightarrow$  Negative  
Calm  $\rightarrow$  ?

## Output

Positive

Here, the model uses several examples to understand the pattern and predict the appropriate output.

## 5 Chain-of-Thought Prompting

Chain-of-thought prompting encourages the model to break a problem into smaller steps before reaching the final answer. It is useful for mathematical and logical problems.

## Prompt

A shop has 10 apples. It sells 3 apples and then buys 5 more.  
Solve the problem step by step and give the final number of apples.

## Output

Initially, the shop has 10 apples.  
After selling 3 apples:  
 $10 - 3 = 7$   
Then, the shop buys 5 more apples:  
 $7 + 5 = 12$   
Final Answer: 12 apples.

The model breaks the problem into smaller steps before arriving at the final answer.

## 6 Role-Based Prompting

In role-based prompting, we tell the AI to act as a particular person or professional. This helps guide the response according to that role's expertise and perspective.

## Prompt

Act as a senior Python developer.  
Explain why using functions is useful in Python.

## Output

Functions help organize Python code into smaller, reusable blocks.  
They make code easier to read, test, debug, and maintain. Instead of writing the same code multiple times, we can define it once inside a function and reuse it whenever needed.

The role gives the model a specific perspective and influences how it explains the topic.

## 7 Instruction-Based Prompting

Instruction-based prompting focuses on giving the model clear and specific instructions. We can specify what the model should do, how it should respond, and what limitations

it should follow.

## Prompt

Explain what an API is.

Use simple language, give 2 real-world examples, and answer in exactly 4 bullet points.

## Output

- An API allows two different software applications to communicate with each other.
- It acts as a bridge between an application and a service.
- Example: A weather app can use a weather API to get current weather data.
- Example: A payment application can use a payment API to process transactions.

Here, the model follows several instructions about the content, language, examples, and output format.

## 8 Conclusion

Different prompting techniques are useful in different situations. Zero-shot prompting works when the task is straightforward and does not require examples. One-shot and few-shot prompting provide examples to help the model understand a pattern. Chain-of-thought prompting is useful for problems that require multiple steps. Role-based prompting gives the model a particular perspective, while instruction-based prompting focuses on clearly describing what the model should do.

Understanding these techniques makes it easier to communicate with LLMs and obtain more useful and consistent responses.